Technical report — Gegenprobe: one schema and a converter for Swiss parliamentary answers

Track 2A — OpenParlData · Online · Team 444 — Roman (+ Claude Code agents, §7) · Evidence site https://gegenprobe.mi11ion.io

maturity	components
ready to apply · ready to deploy	census and Bern repair lists (relinks: one OPD decision, its hash rule) · nightly job (three real nights on standard GitHub runners: one with the venv unit, 10 Oct, and two consecutive with the Docker unit, 11 Oct)
audited · tested prototype	source-adapter lists · rules-only converter
feasibility study · drafts for the eCH group	review mode, whole and virtual affairs · schema v0.1 and v0.2 (P8's roles)

1. Summary

Can one schema hold Swiss parliamentary business? Yes, if each field says which role its value plays (e.g. "answer date" = the government's decision, its letter or the submission date). Tested on OPD's five intervention types: 198,594 of its 325,691 affairs; parliamentary initiatives (5,817) and government business (37,864) are outside the tested scope, and the skeleton does not carry over to them as is (P11, exploratory, 15 affairs: 7 fields fit unchanged, 4 answer fields do not apply, 6 need roles interventions lack). The "Yes" covers the affair's skeleton of 17 fields; the statements inside an answer are a layer of their own (45.9% had no schema account in an inventory over 12 bodies). *Shared:* one schema of 17 fields for communal, cantonal and federal business; every value carries its role and source (page and span, or the OPD record), every gap a typed reason. *Needs roles:* **10 fields** (8 robust + 2 fragile, §5). *Body-specific:* **1 field and 2 constructs**: the submission date (in Vernier, 5/5 affairs, and Lausanne, 3/5, OPD's submission date is printed in neither document; they print the member's signing or lodging date 2 to 28 days earlier); oral answers, councils that publish only minutes. *Untested:* **6 fields**, defined once in v0.1.

Delivered: repairs for 9,580 answers keyed on OPD ids, every one from model-free rules; schema v0.2 with the roles; a converter that gives every value its page and span; a nightly job. Apertus only picks values printed in the document, and code locates them (70B weights via the Public AI gateway; local 8B GGUF on CPU). **Runtime external services:** none by default (public sources only); Apertus review mode sends document text to the Public AI gateway, or nothing with the local 8B. **Cost:** the rules take about 0.5 CPU-hours for all 902,754 documents in OPD's export, about 5 at ten times OPD's volume; review mode USD 0.59 per 1,000 residue documents on the 70B (§3).

92.1%	of interventions with a recorded government answer would have answer text once the proposed repairs are applied, up from 81.6% today: 91.5% from row-checked repairs + 0.6 points from the 592 agenda joins and 39 review rows
9,580	answers with a model-free repair proposed to OPD, keyed on OPD ids; the Bern plan ran on PostgreSQL: 2,975 rows in one transaction that rolls back on any deviation
10 of 17	fields are shared once their roles are explicit (8 robust + 2 fragile; 6 from the role test, P8, registered, and 4 from OPD's label vocabularies, post hoc); 1 is body-specific; 6 are not yet tested for meaning
186	v0.2 records from 26 parliaments validate: the converter's 120 (105 cantonal and communal + 15 federal, 23 parliaments) and P8's 66 gold records (11 parliaments)

Apertus, where nobody wrote rules	what Apertus adds	what the model-free baseline reaches
24/29 answer dates on new layouts (the date test, P2; 15 parliaments)	exact on 53/62 documents against the baseline's 44/62: right on 10 where the baseline is wrong, the reverse on 1 (exact McNemar p = 0.0117; parliament-resampled interval of the difference [0.056, 0.243]; post hoc); 4 of its 7 wins on answered documents are documents OPD types as interventions, where that baseline returns nothing by construction	the latest date on answer-typed documents: 18/29 answer dates, 44/62 exact; registered: the rules 2/29
50/141 author roles in 14 parliaments (the author test, P10)	Apertus adds precision deciding who is an author (+7.3 points, post hoc)	Rules 17/141; most of the gain comes from the model-free candidate finder (document order 52/141)

Sources: 92.1% and 9,580 from the census and the repair lists (no model); 186 from `results/schema/v0_2/validation_all.json` (the 15 federal records from an exploratory cell); 24/29 from P2, registered against the rules (the baselines post hoc, proposed in review; a generic keyword rule finds 16/29, exact McNemar p = 0.057); 50/141 from P10, registered (recall +23.4 points over the rules, Holm p = 0.032), level with OPD's contributor list (37/141; H3a, Holm p = 0.17); the author filter's +7.3 points: one-sided p = 0.012, uncorrected.

Glossary. **P1–P8, P10**: the ten tests (P3b replicates P3), each frozen by hash before its labels (§6). **D-numbers**: logged deviations from a registration. **Branch B**: P1's registered outcome that keeps Apertus off the unattended path. **Post hoc**: analysed after the labels, often proposed in review; **registered**: decided by the frozen rule.

1. **Schema.** P8 finds roles in authorship, co-signatory lists, the instrument behind a harmonised type and the language (text, OPD's tag and the affair's can differ). The converter fills v0.2's roles with fixed rules (75 of its 105 records carry printed authors with their role); on 5 parliaments its rules never saw, the language is right on 60/60 documents and author (name, role) pairs reach precision 39/49 at recall 39/137 (exploratory).
2. **Ready for OPD.** Bern's 2,421 ready relinks and 48 ready fetches, from a linking bug OPD confirmed (correspondence, 10 Oct 2026); 554 `begin_date` corrections; lists from 50 model-free source adapters (blind AI annotators: 185/186 counted rows are the executive's answer, post hoc re-read); 9 OPD vocabulary labels wrong as published; a Fedlex resolver for 29,946 federal placeholder documents (§5).
3. **Apertus** earns its place choosing the right value among printed candidates (table above; by parliament the registered P2 gain over the rules is [-0.02, 0.54]); on the review queue 95.8% of its promotions are answers whose printed answer date passes the date check, 73.2% for a no-model exact-date match (P7-H4, registered, weighted). *Who is an author* (P10, registered: recall +23.4 points over the rules, precision non-inferior; not significantly better than OPD's list, H3a). Where rules exist it adds about one document (P1: cascade 139/154, model-free cascade 138/154, post hoc). In Italian (P11, exploratory) the pointer is exact on 12/12 answers [75.7, 100], the rules on 8/12.

2. Architecture

Census (`census/`) streams the OPD exports once into a per-affair *answer-link status*, the feasibility matrix and a comparability measure; a derived layer (`census/corrected.py`) places every registered gap row once the repair routes have looked (Fig. 1). **Repair** (`repair/`): deterministic routes per cause, each row checked (affair number or title, answer heading, date against OPD's answer event), as lists keyed on OPD ids; a nightly job (`deploy/nightly/`) re-runs the Bern route. **Converter** (`convert/`, `qa/`): OPD metadata and events, then frozen DE/FR/IT anchor rules, then the Apertus closed-set pointer on the residue; modes `rules_only` (default, no model) and `review` (suggestions with page, span and margin, flagged `needs_review`). **Schema and validator** (`schema/`, `validate/`): Pydantic schema v0.1, normative; a value carries its role, page, span, quote hash, route and rule, a null its typed reason (`data/schema/null_reasons.json`); `QAItem` pairs question *n* with its answer passage.

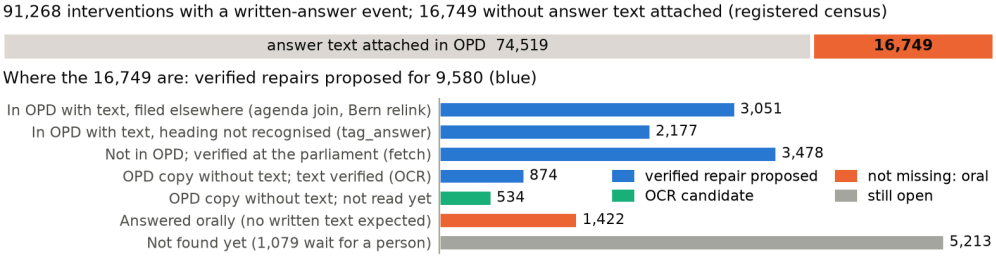


Figure 1. The registered 16,749 placed once each (`results/census/corrected.json`). Still to find: 5,213 (1,079 of them already waiting for a person).

3. Use of Apertus

Closed form only. Apertus never writes a value. (i) *Verifier* (P7, `repair/adapters/adjudicate.py`): for an adapter review row it answers "does this document print the executive's answer to this intervention (number, title)?" (yes vs no scored after a JSON prefix, $\theta_A = 8$) and picks the answer date among the printed dates or none ($\theta_B = 10$); a row is promoted only if both pass and the date passes the adapters' own date rule. (ii) *Pointer* (`convert/pointer.py`, prompts `src/prompts/{de,fr,it}`): the document (first 12,000 characters) and the question in the user message, the assistant turn prefilled with a JSON prefix, `null` and every distinct printed date scored by summed log-probability; null if `lp(null) − logsumexp(lp(dates)) > θ`. Code locates every span (`pointer.locate`). θ does not transfer across builds, so it is fitted per build on dev documents and frozen (θ = 10.40, AUROC 0.99). **Why closed:** asked freely, Apertus dates 15/15 (8B) and 14/15 (70B) Bern interventions that have no answer.

Which model. Every committed gateway response (17,568) comes from one build, Apertus 70B by inference (route and label `swiss-ai/apertus-v1.5-8b`, fingerprint `…tp8-pp2-dd237840`; check in docs/evaluation.md); the response `model` is checked on every call. On 12 Oct 2026 Public AI switched the route to another build (label `swiss-ai/Apertus-v1.5-70B`, fingerprint `…tp2-bfd70257`, other log-probabilities); the build guard caught it live

and stopped P11's role reader, so no result mixes builds, and review mode refuses the new build (exit 3) until its θ is fitted. The **8B** (`swiss-ai/Apertus-v1.5-8B`, a community 4-bit GGUF, sha256-pinned) runs locally on CPU with its own θ = 16.73 (exploratory): on P2's unseen parliaments the registered P2-H2 test over the rules gives 13 vs 0; its two-stage reader v2 cuts dates on answerless documents from 18/34 to 5/34 (P6). As P7's review-queue verifier (exploratory; 88 of the 190 annotated rows scored before the deadline) no θ meets the dev rule, so the 8B promotes nothing. **Recommendation:** `rules_only` unattended; local 8B: date suggestions only, where no text may leave the machine; the review-queue verifier needs the 70B, for accuracy on infrastructure the operator chooses (bf16 weights about 140 GB; operator sheet, `docs/runbook.md`). Review mode stops (exit 3) on a served build θ was not fitted on.

4. Data and operations

Data. OPD bulk exports (snapshot 6 Oct 2026; CC BY 4.0, "Source: OpenParlData.ch"), Bern open data, Fedlex, parliament sites, fetched politely (≥ 1.2 s per host and process). `data/` and `results/` are CDLA-Permissive-2.0 and carry OPD's attribution. **Sovereignty and data protection.** No third-party processor by default; besides the public sources, Hugging Face once for the pinned 8B, and a review-model endpoint (Public AI, or the local 8B, where no text leaves the machine) and an alert relay only if the operator sets them. The operator sheet's nDSG row names the personal data (office holders named in public documents), the route that discloses text abroad (Public AI) and the cache retention (default 30 days); for public bodies it recommends rules only or the local 8B. **Operations.** `make run` and the nightly unit run the image read-only, as a non-root uid, with `--cap-drop ALL`; A network night refuses to start without a contact URL in `GP_USER_AGENT`. **robots.txt:** we found, audited and fixed our own robots.txt bug; every fetch path now fails closed (disclosure: `docs/incidents.md`).

5. Results without a model: schema verdict, census, repair

Table 1. The 17 fields. *Class*: role split = one label covers several roles, measured on OPD's labels (`results/census/comparability.json`) or on the documents (P8, `results/p8/`); fragile = not under both P8 checks; shared = one v0.1 definition. *Source* of the split (committed results only): practice (parliaments differ; the standard encodes the role), OPD (likely a data or tagging error OPD can correct), both, or open (the labels cannot tell). *Where it lives*: parliaments where OPD metadata carries it · carries it for part of the affairs · only the PDF · not in OPD's data (`results/census/matrix.json`).

field	class	source	evidence	where it lives (meta · mixed · PDF · none)
answer date	role split	both	decided, dispatched or registered; in 7 of 54 bodies the submission date	**metadata**: 53 · 1 · 0 · 0
procedural history	role split	open	35 event labels, several date roles	**metadata**: 78 · 9 · 1 · 1
type	role split	both	42 of 1,038 raw labels map to two types; P8: the instrument	partly: 61 · 24 · 2 · 2
procedural state	role split	open	32 labels ambiguous	partly: 41 · 27 · 18 · 3
decision outcome	role split	practice	3 labels ambiguous	partly: 7 · 31 · 33 · 6
submitter	role split	practice; OPD in GR	P8: sole, joint or group author	partly: 23 · 61 · 5 · 0
co-signatories	role split	practice	P8: includes authors of record	PDF only: 0 · 2 · 72 · 12
language	role split	both	P8: tag names the other language; text, tag and affair differ	partly: 14 · 53 · 20 · 2
addressed body	split, fragile	practice	P8: the council (Bellinzona)	partly: 12 · 10 · 54 · 10
government recommendation	split, fragile	both	P8: by stage	partly: 3 · 8 · 56 · 10
submission date	body-specific in 2	OPD	P8: OPD's date printed nowhere (Vernier 5/5, Lausanne 3/5); a data-source split	**metadata**: 89 · 0 · 0 · 0
answer-link status	shared	—	defined once in v0.1	**metadata**: 54 · 0 · 0 · 0
decision date	shared	—	defined once in v0.1	partly: 7 · 30 · 34 · 6
Q/A pairing	shared	—	defined once in v0.1	partly: 22 · 25 · 4 · 3
usable text	shared	—	defined once in v0.1	partly: 41 · 42 · 0 · 6
dispatch date	shared	—	defined once in v0.1	PDF only: 0 · 0 · 47 · 7
related affairs	shared	—	defined once in v0.1	PDF only: 0 · 0 · 76 · 13

Of the 10 role splits, 6 come from the role test (P8, registered) and 4 from OPD's label vocabularies (post hoc); by source of the split 4 parliamentary practice, 4 partly OPD's tagging (which OPD can correct), 2 open. *Do v0.2's roles cover the values?* Every value of the six role-split fields falls into a named v0.2 role: 302/302 across 58 field × parliament cells (post hoc; an exhaustiveness check, since v0.2's roles were written from P8's

categories); in P10's gold from 15 parliaments none of the 197 people needed the guide's open `other` role (0/197, post hoc). 4 fields are in OPD's metadata almost everywhere, 13 partly or only in the PDF. Two further *concepts* are body-specific: oral answers in question time (23.6% of question-time answers) and councils that publish only meeting documents (16 bodies; v0.1 marks affairs rebuilt from minutes `virtual`, P4).

Cross-parliamentary question. Of the 91,268 interventions with a recorded government answer, 81.6% have answer text attached in OPD (the registered count); 82.3% also count the 592 answers OPD files under an agenda item of the affair (not sample-checked); 85.6% with the repairs from what OPD already holds, 88.3% after the Bern relinks, 92.1% with the checked fetches (`results/report/r9_numbers.json`).

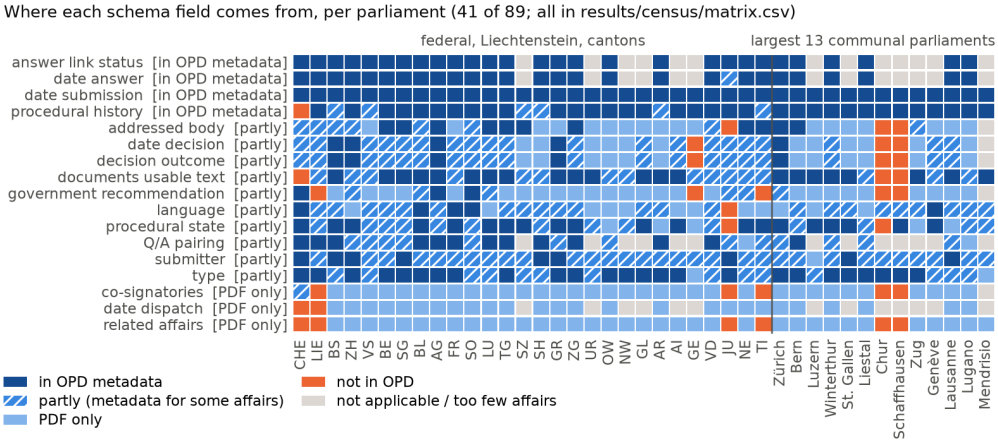


Figure 2. Where each field comes from, per parliament. Orange = not in OPD's data for that parliament (the parliament may still publish it).

Repair. *Bern*: of 3,532 answers the canton publishes OPD attaches 229 (6.5%), although 96.1% are in OPD and linkable without a model. 2,459 relink candidates = 2,421 that pass every check against OPD's own copy (affair number, heading, decision date = OPD's answer event [97.9, 98.9]) + 38 to review. The document `hash` is sha256(body_key + parent_type + url) (49,810 of 49,810 rows), so relinking changes each identifier and file URL unless OPD keeps the old one; the plan carries both values. *Source adapters* (`docs/adapters.md`): readers for 50 parliaments (27 yield counted rows) verify 6,473 answers by affair number or title, the parliament's answer formula and a date check; Valais's 140 date-window rows moved to review: OPD's Valais answer event is the session that debated the motion, after the government's letter (logged in `results/repair/adapters/retier/retier_r18.json`). 2,177 of them are already in OPD with text under a heading the census lexicon does not list: the repair is a category on the document (`tag_answer`). **Blind audit v3** (`results/repair/adapters/audit3/`; 60 decoys, key hashed on GitHub first): on the counted rows Claude Opus judged 185/186 the executive's answer (3 dates judged wrong, all Solothurn, where the auditor saw the committee's "Beantwortet" date) and Qwen-SEA-LION 186/186 (post hoc re-read over the strata that still count; as run on 200 rows: 197/200, 12 of its 15 date verdicts on the 14 sampled Valais date-window rows).

6. Evaluation: ten pre-registrations

Design. Each test froze code, prompts and θ by sha256 before any label was committed (P1's hypothesis script followed its labels, D4). Unit (P1, P2, P6): one document's answer date, strict. Arms on identical inputs: **rules**; **pointer**; **cascade** (rules, then the pointer on the residue); **naive** free generation, plain and null-allowed; trivial baselines. **Gold, no human labelling by design:** two blind agent annotators (Claude Opus, Sonnet), a third breaks ties, quotes verified by code, raw files committed; all 96 annotator sessions (P1–P8, P10, the document-parts check, the federal cell) ended before GitHub received their labels (`results/eval/annotation_sessions.json`).

Table 2. The ten tests, in registration order.

test	n	what it tests	registered result	note
P1	154 documents, parliaments with rules	answer date: rules vs Apertus	**partial**: H3 held; H1 null safety failed (7/90); H2 Branch B	rules alone 135/154, so no model by default
P2	62 documents, 15 parliaments without rules	rules transfer; pointer vs rules	**3 of 4 held**; H3 null safety failed (4/33)	pointer 24/29 dates, rules 2/29
P3	29 whole affairs, 10 parliaments	7 fields: metadata, rules, Apertus	**partial**: no field unattended; Apertus recall held, precision below	feasibility study (D3-5)
P3b	36 fresh affairs, 9 parliaments	replication of P3	**held**: replicated (S1, S2)	S2 lenient; type 35/36; addressed body 2/33
P4	120 agenda items, 6 sittings	affairs rebuilt from minutes	**partial**: H2, H4 held; H1, H3 failed	recall 99/120, precision 99/99

test	n	what it tests	registered result	note
P5	40 fresh affairs, 10 parliaments	Q/A pairs, people, recommendation	**partial** tier (H1); H2, H5 held; H4 as rules vs free generation; H3 no gain	*not yet* when parliaments are resampled
P6	68 fresh documents, 8 parliaments	answer-date reader v2 vs free prompting	**no hypothesis held**	free prompt 62/68, v2 59/68
P7	478 review rows, 190 blind packets	Apertus as verifier of the review queue	**H1 holds** (120/120); H4 supported; H3 not supported	214 of 917 rows promoted
P8	66 fresh affairs, 11 parliaments	meaning of 7 fields in the documents	**none uniform**: 6 role split, 1 body-specific	5 robust; meaning measured for 11 of 17
P10	34 fresh affairs, 14 parliaments without rules	author roles: Apertus reader vs rules, OPD's list	**3 of 4 held**: H1, H2, H3b; H3a not	pairs right 50/141, rules 17/141, OPD 37/141

Unattended (95% lower bound ≥ 0.80, document-level intervals): the rules' answer date where rules exist (P1, helped by right nulls: dates found 47/64), the type on whole affairs (P3b, 35/36), and in P5 question detection (recall 167/187, precision 167/175), the reasoning span (37/40) and co-signatory precision (101/103). **Cross-checks.** `Qwen-SEA-LION-v4-32B-IT` (AI Singapore) relabelled P1, P2, P6 and P7 blind: date κ 0.80, 0.87 and 0.86, P7's is-answer labels 185/190; no registered decision changes. Non-LLM checks: OPD's event matches 37 of 44 checkable P1 dates; P6: 10/10, every gold quote on its page (34/34). Holm over all 20 registered confirmatory tests of P1–P7: the 9 that held survive (`results/eval/posthoc_r8_multiplicity.json`); whole parliaments are resampled as well.

P1 arm (154 documents)	exact [95% CI]	answered: found /64	wrong printed date	invented	answerless dated /90
rules only (default)	**135/154 [81.5, 92]**	47	4	0	2
pointer	130/154 [77.9, 89.3]	46	16	0	6
cascade	139/154 [84.6, 94]	56	11	0	7
naive, null-allowed	134/154 [80.8, 91.4]	47	11	0	3
trivial: latest date, answer-typed docs	*131/154 [78.6, 89.8]*	*44*	*16*	*0*	*3*
model-free cascade (post hoc)	*138/154 [83.8, 93.5]*	*52*	*11*	*0*	*4*

P1: the rules (135) and the latest date on answer-typed documents (131) are level (paired p = 0.48, post hoc; P1's rules entered with its sample, D12), so P2 is the test of transfer. H1 null safety failed (the cascade dates 7 of 90 answerless documents, 6 of them council reports; upper bound 15.2% vs 10%); H2 gave Branch B (cascade alone right on 5, rules alone on 1, p = 0.22), so Apertus left the unattended path for an opt-in review suggestion (D6); H3 held (AUROC 0.969).

P2 arm (62 documents)	exact [CI]	answered: found /29 [CI]	wrong date	invented	answerless dated /33
rules only	35/62 [44.1, 68.1]	**2/29** [1.9, 22]	2	0	0
pointer	**53/62** [74.7, 92.2]	**24/29** [65.5, 92.4]	8	0	4
naive, null-allowed, re-parsed (D2-4)	54/62	24/29	7	1	3
trivial: latest printed date	*24/62*	*22/29*	*38*	*0*	*31*
trivial: latest date, answer-typed docs	*44/62*	*18/29*	*13*	*0*	*7*

P2: the rules collapse off their home layouts (47/64 → 2/29). H2 held: on all 62 documents the pointer beats the rules on 22 and loses on 4 ($p = 5.3 \times 10^{-4}$); resampling the 15 parliaments, the gain is [-0.02, 0.54], so it sits in a few parliaments. "Answerless" means no qualifying answer date: the pointer's 4 such dates fall on documents with an answer but no qualifying date; of the 25 with no answer at all it dates none (post hoc; with P6's v2 none of 53 across P2 and P6; P6's v1 dated 1 of 28). *Post hoc, proposed in review:* a generic DE/FR/IT keyword rule is exact on 45/62 and finds 16/29 answer dates (pointer vs rule: McNemar p = 0.077; on answer dates 0.057).

P3/P3b/P4. Default (metadata + rules) on 29 affairs: type 27/29, answer date 22/27, 27/27 with Apertus suggestions. P3b replicated by its registered criteria; answer date 20/35 by default (C), 31/35 with Apertus suggestions. P4: 99 of 120 agenda items as virtual affairs, precision 99/99, no decision invented (0/36). **P5.** Rules-only pairing: precision 118/143 = 0.83, recall 118/176 = 0.67, *partial* with document-level intervals, *not yet* when parliaments are resampled (precision [0.66, 0.97]); Recommendation 34/40 by rules; submitter rules 12/40, pointer 24/40, free generation 34/40.

P6 — reader v2 (an answer-present gate, then the gold guide's decision-before-letter-date rule). No registered hypothesis held: v2 vs free prompting 1:4 (p = 0.38); v2 dates 5/34 answerless documents (upper bound 30.1% vs 15%), all 5 containing an answer without a qualifying date; v2 vs v1 5:2 (p = 0.45).

P7 — Apertus as a precision verifier of the review queue (`eval/prereg8/`, LOCK8 before the draw). Test frame: 478 eligible review rows outside every audit; 190 blind packets, gold A1 Opus / A2 Sonnet. H1 (primary): promotion precision 120/120, Wilson lower bound 96.9% ≥ 0.90: **holds**. H4: dated precision 95.8% vs 73.2% for a trivial exact-date match, +22.7 points [10.2, 34.3], weighted, as registered (unweighted +7.5): **supported**. H3: the rows free generation with a printed-date filter adds are 28/30 right: **not supported**. Base rate: 86.7% of

eligible queue rows are answers, but only 40.9% are answers whose own date passes the rule, so the gain is the date check. *Post hoc* (`results/p7/`): the metric measures how often promotions are date-checkable, not the date picked. Apertus's chosen date is the gold date on 113/120 of its promotions, the exact-date match's on 127/146 (its promotions are answers on 146/146), each on its own promotions (not paired). A model-free rule accepting any printed date of the same closed set recovers more rows with less precision (76.2% dated; 369 vs 214 queue rows).

P8 — what the shared fields mean (`eval/prereg9/`, LOCK9 before the draw). Blind labels of what seven fields denote in the two documents OPD links. Share in v0.1's meaning, pooled (parliament-cluster 95% interval): submitter 35/65 [33.3%, 75.0%], co-signatories 10/41 [6.8%, 46.3%], submission date 44/57 [57.1%, 91.8%], type 55/64 [76.2%, 95.3%], language 34/36 [83.3%, 100.0%], each verdict unchanged on agreed labels only and on Qwen's labels; addressed body (57/60) and recommendation (31/34) are fragile.

P10 — Apertus reads author roles where no rules exist (`eval/prereg10/`, LOCK10 before the draw and the labels; D1). 34 fresh German and French affairs (of 38 planned; French 4 short under the registered fill rule), 14 parliaments outside every rule set, 141 gold (name, role) pairs (A1 Opus / A2 Sonnet, pair F1 0.868). Pairs right / printed: Apertus 50/96 (precision 0.52, recall 0.35, F1 0.42), rules 17/55 (0.31, 0.12, 0.17), OPD's contributor list as first author and co-signers 37/104 (0.36, 0.26, 0.30). Confirmed: H1, recall over the rules, +23.4 points [5.9, 49.0], Holm p = 0.032; H2, precision non-inferior (+0.21); H3b, over the same names with a trivial role. Not confirmed: H3a, over OPD's list (+0.12, Holm p = 0.17): where OPD lists the authors, its order is a strong baseline. *Post hoc, proposed in review:* the same candidates in document order with OPD's registered mapping match 52/141 (111 printed; 0.47, 0.37, F1 0.41). The gain over the rules is the candidate finder plus document order (H3b's registered baseline reads every candidate as a joint author: 16/141); Apertus's author filter, on the same order rule, keeps recall and adds +7.3 points precision [0.8, 18.2] (uncorrected).

Federal cell (exploratory, `eval/fed/`, LOCKFED before draw and labels): 15 Federal Council answers print no answer date and no author; rules, pointer and converter date none (15/15 right nulls), the latest printed date dates an answerless document on 6/15; all 15 v0.2 records validate.

P11 (exploratory, `eval/p11/`, LOCKP11 before draw and labels; D1). *Italian*, 12 interventions from Ticino, Mendrisio, Lugano, Locarno and Bellinzona (12 of the 24 documents read from OPD's copy where robots.txt disallows the source): the pointer is exact on 12/12 [75.7, 100] (10 printed dates, 2 right nulls, none invented), the latest date on the answer document on 9/12 (it dates both answerless affairs), the rules on 8/12 (in-sample in Ticino, Mendrisio, Lugano); author roles (70 gold pairs): rules precision 0.875 at recall 0.10, the converter's record F1 0.30, document order and OPD's list 0.22, the role reader 0.12 on the 5 affairs it read before the build switch. *Outside interventions*, 15 government-business and initiative affairs: 7 of 17 fields carry over, 4 answer fields do not apply, 6 need new roles, chiefly the executive acting as proposer: adopting, endorsing and signing its own proposal (government business 7 of 8 affairs, initiatives 2 of 7); the converter returns a valid v0.2 record for 3 of 15.

What the tests say. *Provenance by construction:* no arm that picks from printed dates returned a date printed nowhere (P1, P2, P6), with a page, span and margin on every value. **Null safety** is left to the reviewer: the registered null-safety tests on the 70B failed (P1-H1, P2-H3, P6-H2), which is why the default runs no model.

7. Next steps, reproducibility, declarations

Next steps. Next come the contents of the document parts marked by page and image detection (page precision 4/8). With OPD: its hash decision for the relinks and the 30-row spot-check kit (`results/opd_spotcheck/`); with the eCH group: `QAItem` and the five gaps. Author rules are per layout (held out: pair recall 39/137); where none exists the candidate finder, which offers 87 of the 141 gold names there, is the next lever. *Limits of the answer, each with its next test:* three splits rest on one parliament each, addressed body (Bellinzona 2/4), recommendation (Vernier 0/3) and language (Fribourg 4/6, OPD's tag), and on the affairs the gold and the Qwen check label alike the first two become shared (5/7 verdicts hold; post hoc): more parliaments per field. Italian rests on P11's exploratory cell (12 affairs; author-role F1 0.30 or lower for every arm; the role reader stopped at 5): a registered Italian test, with θ and the role reader refitted for the new gateway build. Outside interventions: v0.3 roles for the executive as proposer and a converter route for affairs without a member's text. 6 fields are shared by definition only: their meaning test. **Reproducibility.** `make run` (Docker, CPU) recomputes the census and the Bern join, re-derives the P1 rules arm from the source PDFs, replays P1–P8 and P10 (all locks verified), regenerates the summary-number files byte for byte and runs the claims check, which binds every number from 10 up and every scope word in this report to its claim and checks this PDF against its source. **Declarations.** Claude Code agents (Anthropic) wrote most code, documentation and the gold labels. Public AI (may route outside Switzerland) served evaluation only. Report CC-BY-4.0.